

Written HW6

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I use Neovim (BTW)

Full HW repository at
github.com/LoganCTanner/School

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1. Suppose we roll a fair die n times. Let X be the number of 1's and Y be the number of 2's that appear.

- (a) Do you expect that $\text{Cov}(X, Y)$ will be positive, negative, or 0? Explain.

Answer:

Expect to be 0 or negative, 0 because the rolls are independent, negative because if there's a large amount of 1's then 2's must be lower and vice versa.

- (b) Now actually compute $\text{Cov}(X, Y)$.

$$\begin{aligned} E[XY] - E[X]E[Y] \\ E[X] = E[Y] = n \times \frac{1}{6} = \frac{n}{6} \\ E[X]E[Y] = \frac{n^2}{36} \\ E[XY] = E[X_1Y_1 + \dots + X_nY_n] \\ \Rightarrow E[X_iY_i] = 0 \quad \forall i \in [1, n] \\ \text{(cannot be both 1 and 2 on the same roll)} \\ \Rightarrow E[X_iY_j] = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36} \quad \text{s.t. } i \neq j \\ E[XY] = 0 + 2 \cdot \binom{n}{2} \frac{1}{36} = \frac{n(n-1)}{36} \\ E[XY] - E[X]E[Y] = \frac{n^2 - n^2 - n}{36} = -\frac{n}{36} \end{aligned}$$

- (c) Find $\rho_{X,Y}$. Is there anything surprising to you about your answer?

$$\begin{aligned} \sigma_X = \sigma_Y \\ \text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i) \\ E[X_i^2] - E[X_i]^2 = \frac{1}{6} - \frac{1}{36} = \frac{5}{36} \\ \text{Var}(X) = n \cdot \frac{5}{36} \Rightarrow \sigma_X = \frac{\sqrt{5n}}{6} \\ \rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y} = -\frac{n}{36} \cdot \frac{36}{5n} = -\frac{1}{5} \end{aligned}$$

I would have assumed beforehand that it would have been something not so clean.

2. Recall the following scenario from Homework 3:

A restaurant offers kids' meals that come with a toy. There are 5 different types of toy available, and each is equally likely to come with a given meal. Find the expected value and standard deviation of the number of types of toy still uncollected after 7 meals have been purchased.

- (a) Answer the question again, this time using what we have learned about covariance to find the std. dev.

$$\text{Var}(X) = \text{Cov}(X, X) = \text{Cov}\left(\sum_{i=1}^5 X_i, \sum_{j=1}^5 X_j\right)$$

$$\text{Cov}\left(\sum_{i=1}^5 X_i, \sum_{j=1}^5 X_j\right) = \sum_{i=1}^5 \sum_{j=1}^5 (E[X_i X_j] - E[X_i]E[X_j])$$

For all $i \neq j$:

$$E[X_i X_j] = \left(\frac{3}{5}\right)^7$$

$$E[X_i] = E[X_j] = \left(\frac{3}{5}\right)^7$$

$$\left(\frac{3}{5}\right)^7 - \frac{16^7}{25^7} = \frac{15^7 - 16^7}{25^7}$$

For all $i = j$:

$$E[X_i^2] - E[X_i]^2 = \left(\frac{4}{5}\right)^7 - \left(\frac{16}{25}\right)^7 = \frac{20^7 - 16^7}{25^7}$$

$$\text{Cov}\left(\sum_{i=1}^5 X_i, \sum_{j=1}^5 X_j\right) = \frac{20^7 - 16^7}{25^7} + \frac{15^7 - 16^7}{25^7}$$

$$\sigma_X = \sqrt{\frac{20^7 - 16^7}{25^7} + \frac{15^7 - 16^7}{25^7}}$$

- (b) Compare your work in (a) with the posted solution to Homework 3 Problem 4.

Answer:

Told to not complete since answer wasn't posted.

3. Suppose we roll a fair die twice, and let X be the number of 1's and Y be the number of 2's that appear; i.e., we are returning to the situation in Problem 1 but fixing $n = 2$.

- (a) What is $M_X(t)$, the mgf of X ? What is $M_Y(t)$? You may use a result from class as long as you explain why it applies.

Answer:

$$\begin{aligned} M_X(t) &= E[e^{Xt}] = E[e^{0t}] + E[e^{1t}] + E[e^{2t}] \\ &= e^0 P(X=0) + e^t P(X=1) + e^{2t} P(X=2) \\ &= \frac{25}{36} + e^t \frac{10}{36} + e^{2t} \frac{1}{36} \end{aligned}$$

Then, since $P(X=i) = P(Y=i)$

$$M_Y(t) = \frac{25}{36} + e^t \frac{10}{36} + e^{2t} \frac{1}{36}$$

- (b) Find $M_{X,Y}(t, u)$, the joint mgf of X and Y .

Answer:

$$\begin{aligned} &\begin{bmatrix} P(X=0 \cap Y=0) & P(X=0 \cap Y=1) & P(X=0 \cap Y=2) \\ P(X=1 \cap Y=0) & P(X=1 \cap Y=1) & \\ P(X=2 \cap Y=0) & & \end{bmatrix} \\ &= \\ &\begin{bmatrix} \frac{16}{36} & \frac{8}{36} & \frac{1}{36} \\ \frac{8}{36} & \frac{2}{36} & \\ \frac{1}{36} & & \end{bmatrix} \end{aligned}$$

$$\Rightarrow \frac{16}{36} + e^u \frac{8}{36} + e^{2u} \frac{1}{36} + e^t \frac{8}{36} + e^{t+u} \frac{2}{36} + e^{2t} \frac{1}{36}$$

- (c) Check that substituting $t = 0$ and $u = 0$ respectively into $M_{X,Y}(t, u)$ results in the marginal mgfs of X and Y .

Answer:

$$\begin{aligned} M_{X,Y}(t, 0) &= \frac{16}{36} + \frac{8}{36} + \frac{1}{36} + e^t \frac{10}{36} + e^{2t} \frac{1}{36} = \frac{25}{36} + e^t \frac{10}{36} + e^{2t} \frac{1}{36} \\ M_{X,Y}(0, u) &= \frac{16}{36} + \frac{8}{36} + \frac{1}{36} + e^u \frac{10}{36} + e^{2u} \frac{1}{36} = \frac{25}{36} + e^u \frac{10}{36} + e^{2u} \frac{1}{36} \end{aligned}$$

- (d) Show that, for *any* joint mgf of two variables, $\left. \frac{\partial^2 M_{X,Y}(t, u)}{\partial t \partial u} \right|_{t=0, u=0} = E[XY]$.

$$\begin{aligned} \frac{\partial^2 M_{X,Y}(t, u)}{\partial t \partial u} &= E\left[\frac{\partial}{\partial u} X e^{Xt+Yu}\right] = E[XY e^{Xt+Yu}] \\ &E[XY e^{0t+0u}] = E[XY] \end{aligned}$$

- (e) Use this formula to (re)compute $\text{Cov}(X, Y)$. Verify that your answer matches 1(b).

$$\begin{aligned}
 \text{Cov}(X, Y) &= \frac{\partial^2 M_{X,Y}(t, u)}{\partial t \partial u} - \frac{\partial M_{X,Y}(t, 0)}{\partial t} \frac{\partial M_{X,Y}(0, u)}{\partial u} \\
 &= \frac{2}{36} e^{0+0} - \left(\frac{10}{36} e^0 + 2 \frac{1}{36} e^0 \right) \left(\frac{10}{36} e^0 + 2 \frac{1}{36} e^0 \right) \\
 &= \frac{2}{36} - \left(\frac{12}{36} \right)^2 \\
 &= \frac{2}{36} - \frac{1}{9} = \frac{2}{36} - \frac{4}{36} \\
 &= -\frac{2}{36} = -\frac{n}{36}
 \end{aligned}$$